WORKED SOLUTION

STEP II 2024, Question 6

Pure • Binomial Series Coefficients

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every part runs on ratios of successive terms. A binomial coefficient $\binom{2n}{n}$ and a binomial-series coefficient are both products that grow by one factor at each step, so comparing the nth term with the $(n-1)$th gives a short recurrence. Two sequences that obey the same recurrence and start at the same value are equal. The final identity then comes from writing $(1-x)^{-3/2}$ as $(1-x)^{-1/2} \times (1-x)^{-1}$ and comparing coefficients, because multiplying by $(1-x)^{-1} = 1 + x + x^2 + \cdots$ turns a sequence of coefficients into its running totals.

Solution

Part (i)

Let $P(n)$ be the statement $T_n = \dfrac{1}{2^{2n}}\binom{2n}{n}$.

Base case. $P(0)$ holds, since $\frac{1}{2^0}\binom{0}{0} = 1 = T_0$.

Inductive step. Suppose $P(n-1)$ holds for some $n \geq 1$. Since $\binom{2n}{n} = \dfrac{(2n)!}{n!\,n!}$ and $\binom{2n-2}{n-1} = \dfrac{(2n-2)!}{(n-1)!\,(n-1)!}$, dividing gives

$$\binom{2n}{n} = \frac{(2n)(2n-1)}{n \cdot n}\binom{2n-2}{n-1} = \frac{2(2n-1)}{n}\binom{2n-2}{n-1}.$$

Then, using the recurrence and $P(n-1)$,

$$\begin{aligned} T_n = \frac{2n-1}{2n}\,T_{n-1} &= \frac{2n-1}{2n}\cdot\frac{1}{2^{2n-2}}\binom{2n-2}{n-1} \\ &= \frac{2n-1}{2n}\cdot\frac{1}{2^{2n-2}}\cdot\frac{n}{2(2n-1)}\binom{2n}{n} \\ &= \frac{1}{4\cdot 2^{2n-2}}\binom{2n}{n} = \frac{1}{2^{2n}}\binom{2n}{n}, \end{aligned}$$

which is $P(n)$. By induction, $P(n)$ holds for all $n = 0, 1, 2, \ldots$.

Part (ii)

By the binomial theorem for a general index, the coefficient of x^r in $(1-x)^{-1/2}$ is

$$a_r = \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{1}{2}-r+1\right)}{r!}\,(-1)^r = \frac{\frac{1}{2}\cdot\frac{3}{2}\cdots\frac{2r-1}{2}}{r!},$$

since each of the r factors in the numerator is negative and the $(-1)^r$ from $(-x)^r$ cancels the signs. Going from a_{r-1} to a_r adds one factor $\frac{2r-1}{2}$ on top and one factor r underneath, so for $r \geq 1$

$$\frac{a_r}{a_{r-1}} = \frac{(2r-1)/2}{r} = \frac{2r-1}{2r}, \qquad \text{that is,} \qquad a_r = \frac{2r-1}{2r}\,a_{r-1}.$$

Ask yourself Why does this recurrence immediately give $a_r = T_r$?

Both sequences start at 1 (because $a_0 = 1$, the constant term, and $T_0 = 1$) and obey the same rule for getting from term $r-1$ to term r. A one-line induction finishes it: if $a_{r-1} = T_{r-1}$, then $a_r = \frac{2r-1}{2r}a_{r-1} = \frac{2r-1}{2r}T_{r-1} = T_r$. So $a_r = T_r$ for all $r \geq 0$, and by (i),

$$a_r = \frac{1}{2^{2r}}\binom{2r}{r}.$$

Part (iii)

In the same way, the coefficient of x^r in $(1-x)^{-3/2}$ is

$$b_r = \frac{\frac{3}{2}\cdot\frac{5}{2}\cdots\frac{2r+1}{2}}{r!}.$$

The numerators of a_r and b_r are both products of r consecutive half-odd numbers. The product for b_r has lost the first factor $\frac{1}{2}$ of a_r and gained a factor $\frac{2r+1}{2}$ at the end. The $r!$ cancels, so

$$\frac{b_r}{a_r} = \frac{(2r+1)/2}{1/2} = 2r+1.$$

(For $r = 0$ both products are empty and $b_0 = a_0 = 1$, which agrees.) Hence

$$b_r = (2r+1)\,a_r = \frac{2r+1}{2^{2r}}\binom{2r}{r}, \qquad r = 0, 1, 2, \ldots.$$

Part (iv)

Since $(1-x)^{-3/2} = (1-x)^{-1/2}(1-x)^{-1}$ and $(1-x)^{-1} = \sum_{s\geq 0} x^s$,

$$\sum_{n=0}^{\infty} b_n x^n = \left(\sum_{r=0}^{\infty} a_r x^r\right)\left(\sum_{s=0}^{\infty} x^s\right).$$

On the right, a term x^n comes from choosing $a_r x^r$ from the first bracket and x^{n-r} from the second, for each $r = 0, 1, \ldots, n$. Each of these choices has coefficient $a_r \cdot 1$, so the coefficient of x^n is $a_0 + a_1 + \cdots + a_n$. Comparing coefficients of x^n on the two sides,

$$b_n = \sum_{r=0}^{n} a_r.$$

Substituting the formulae from (ii) and (iii) gives

$$\frac{2n+1}{2^{2n}}\binom{2n}{n} = \sum_{r=0}^{n} \frac{1}{2^{2r}}\binom{2r}{r},$$

as required. (The working holds for $n = 0$ too, where both sides equal 1.)

Checks

- Small cases: $a_0, a_1, a_2, a_3 = 1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}$, matching $\binom{2r}{r}/4^r = 1, \frac{2}{4}, \frac{6}{16}, \frac{20}{64}$.
- For $n = 3$, the left side of (iv) is $\frac{7\cdot 20}{64} = \frac{35}{16}$, and the right side is $1 + \frac{1}{2} + \frac{3}{8} + \frac{5}{16} = \frac{35}{16}$.
- Directly, $(1-x)^{-3/2} = 1 + \frac{3}{2}x + \frac{15}{8}x^2 + \cdots$, and $b_2 = 5 \cdot \frac{3}{8} = \frac{15}{8}$.
- A computer check with exact fractions confirmed (i)–(iv) for every $n \leq 29$. Summing the first 30 terms of each series at $x = 0.3$ gives 1.19523 and 1.70747, which agree with $(0.7)^{-1/2}$ and $(0.7)^{-3/2}$.

Common slips

- Sign errors in the binomial coefficients: $\binom{-1/2}{r}$ alternates in sign, but the factor $(-1)^r$ from $(-x)^r$ cancels this, so every a_r and b_r is positive.
- Forgetting the base case, or claiming $a_r = T_r$ from the recurrence alone: the starting values $a_0 = T_0 = 1$ must also agree.

Beyond STEP

Multiplying a power series by $(1-x)^{-1}$ to produce partial sums is a basic tool of *generating functions*, which turn questions about sequences into algebra with power series. The numbers $\binom{2n}{n}/4^n$ also appear in probability: this is the chance that a fair coin tossed $2n$ times shows exactly n heads, and Stirling's formula shows it behaves

like $1/\sqrt{\pi n}$ for large n. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit, and Grimmett and Welsh, *Probability: An Introduction*, develops generating functions properly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.